

Student Name:

Reg/ID #:

Convert to IEEE 754 single precision floating point numbers:

1. 333.45

Sol:

$$333 = 101001101$$

$$0.45 \approx 0.01110011...$$

$$333.45 \approx 101001101.01110011$$

$$333.45 \approx 1.0100110101110011 \times 2^8$$

$$S = 0$$

$$\text{Fraction} = 01001101011100110011010$$

$$\text{Exp} = 127 + 8 = 135 = 10000111$$

$$0 \mid 10000111 \mid 01001101011100110011010$$

2. 1.001

Sol:

$$1 = 1$$

$$.001 = 0.00000000010000011000101$$

$$1.001 = 1.00000000010000011000101 \times 2^0$$

$$S = 0$$

$$\text{Fraction} = 0000000000000000000011$$

$$\text{Exp} = 127 + 0 = 01111111$$

$$0 \mid 01111111 \mid 00000000010000011000101$$